

gradbars

Compact data graphics inside LaTeX tables

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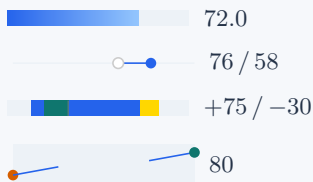
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<https://github.com/Tape5try/gradbars>

The package draws individual data graphics in ordinary table cells or inline text. It provides a common interface for bars, compositions, comparisons, intervals and short trends. Raw values determine geometry; formatting only affects labels.

This English manual contains executable examples with the output above their source. All numbers are illustrative. The Chinese manual provides additional examples and full two-column, longtable, grayscale and Beamer layouts.

A single interface for compact graphics



```
\gradbar{72}
```

```
\graddumbbell[precision=0]{58}{76}
```

```
\gradstack[min=-50,precision=0,stack totals=separate]{60,-20,15,-10}
```

```
\gradspark[spark range=fixed,min=0,max=100,precision=0]{25,45,NA,60,80}
```

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1 Installation and engine requirements

Place `gradbars.sty` beside your main source, or install it in a local TEXMF tree under `tex/latex/gradbars`. Load it with `\usepackage{gradbars}`. Dependencies are TikZ/PGF, xparse, expl3 and collcell (including array). No shell escape or external data-conversion program is required.

The package has no inherent XeLaTeX requirement. PDF output with pdfLaTeX, XeLaTeX and LuaLaTeX is tested, including CSV input. The Chinese manual uses XeLaTeX with ctex and Fandol fonts. This English manual can be compiled with pdfLaTeX. The package itself does not load fontspec or ctex. Use an appropriate font and encoding setup for non-ASCII text in your own documents.

```
\documentclass{article}
\usepackage{booktabs,gradbars}
\begin{document}
\gradbarssetup{width=30mm,max=100,precision=1}
\begin{tabular}{lc}
\toprule
Model & Accuracy (\%) \\
\midrule
Baseline & \gradbar{72.4} \\
Proposed & \gradbar{92.7} \\
\bottomrule
\end{tabular}
\end{document}
```

No `tikzpicture` wrapper is needed. Commands may also be used inside an existing TikZ picture. Settings and named styles follow ordinary TeX grouping. Options are applied left to right: put a theme or preset before individual overrides.

2 Choosing a graphic

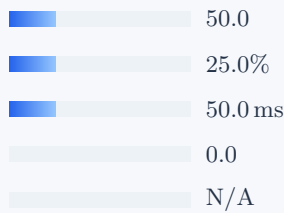
Question	Interface
How large is a value?	<code>\gradbar {value}</code> ; use a negative min for signed values.
How do two values compare?	<code>\gradcompare {reference}{current}</code> for layered bars; <code>\graddumbbell {reference}{current}</code> for connected points.
What are the bounds?	<code>\gradrange {lower}{upper}</code> for supplied endpoints.
What makes up the total?	<code>\gradstack {list}</code> for additive, common-unit contributions.
How does a series change?	<code>\gradspark {list}</code> for equally spaced observations.
What is the uncertainty?	<code>error minus</code> and <code>error plus</code> on a point estimate.

3 Values, scales, labels and missing observations

Defaults are `min=0` and `max=100`. The maximum must be positive and the minimum must be nonpositive. Decimal literals and macros expanding to decimals are accepted; expressions, thousands separators, units in input, and scientific input notation are not. Scientific notation is available as an output format.

Keep `min`, `max` and `width` identical within each comparison column. Zero is not missing. Empty input or uppercase NA denotes missing data for a single observation; missing stack segments are rejected. Overflow warns and clips geometry while preserving the original label. Use `overflow=error` for strict checking.

Raw values, calculated percentages and missing values



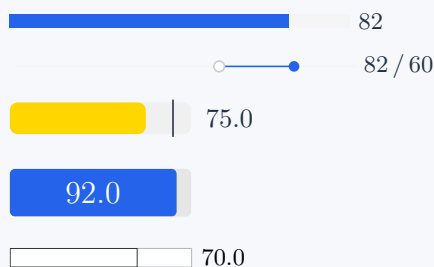
```
\gradbar[max=200]{50}
\gradbar[max=200,value format=percent]{50}
\gradbar[max=200,unit={\,ms}]{50}
\gradbar{0}
\gradbar[missing text={N/A}]{NA}
```

unit appends text. **value format=percent** computes $100v/\text{max}$, ignores unit, and requires **min=0**. For values already expressed as percentages, use **unit={\%}**. **text** overrides the complete label without changing geometry. **precision** accepts integers from 0 to 6. **number format** is fixed, grouped or scientific. Missing labels do not acquire a unit or percent suffix.

4 Appearance, reusable styles and palettes

Seven themes are available: blue, teal, solid, gray, lbyellow, vibblue and cyblu. Nine predefined solid colors are lightgreen, lightyellow, lightblue, lightred, rose, skyblue, gold, lavender and peach. Use **color** for solid fill and **left color/right color** for a gradient. Standard xcolor expressions work.

A reusable layout style



```
\gradbarsstyle{paperarrow}{preset=paper,width=45mm,max=100,precision=0}
\gradbar[style=paperarrow]{82}

\graddumbbell[style=paperarrow]{60}{82}

\gradbar[preset=report,target=90,thresholds={60,80}]{75}

\gradbar[preset=presentation]{92}

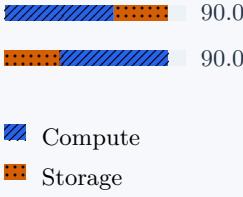
\gradbar[preset=outline]{70}
```

The paper preset uses thin solid bars, a faint track and outside values. Report uses thicker rounded bars; supply actual target and threshold values yourself. Presentation uses larger type and automatic inside placement. Outline uses hollow shapes with fine strokes. Presets do not invent data or scale limits.

Group palettes are categorical (six distinct colors), sequential (five blue levels), diverging (orange through

neutral to teal), and mono (gray with textures). These palettes select by segment position, not by value; short lists cycle. Named categories instead preserve identity across reordered or subsetting data:

Stable category identity and a shared legend



```

\gradbarcategory{Compute}{gradbarsBlue}{diagonal}
\gradbarcategory{Storage}{gradbarsOrange}{dots}
\gradstack[stack names={Compute,Storage}]{60,30}

\gradstack[stack names={Storage,Compute}]{30,60}

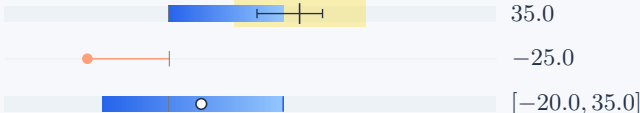
\gradbarslegend[width=50mm]{Compute,Storage}

```

Registered names override positional colors and patterns; unregistered names fall back to the current palette. Definitions are scoped. In print/mono mode, category colors use gray while registered patterns remain. Legends use the same mapping. Patterns are none, diagonal, reverse, dots, crosshatch, horizontal and vertical. Use names or textures alongside color when category distinctions matter.

5 Signed bars, targets, intervals and uncertainty

A signed point estimate and asymmetric uncertainty



```

\gradbar[min=-50,max=100,width=65mm,
  target=40,band={20,60},error minus=8,error plus=12]{35}

\gradbar[min=-50,max=100,width=65mm,shape=lollipop]{-25}

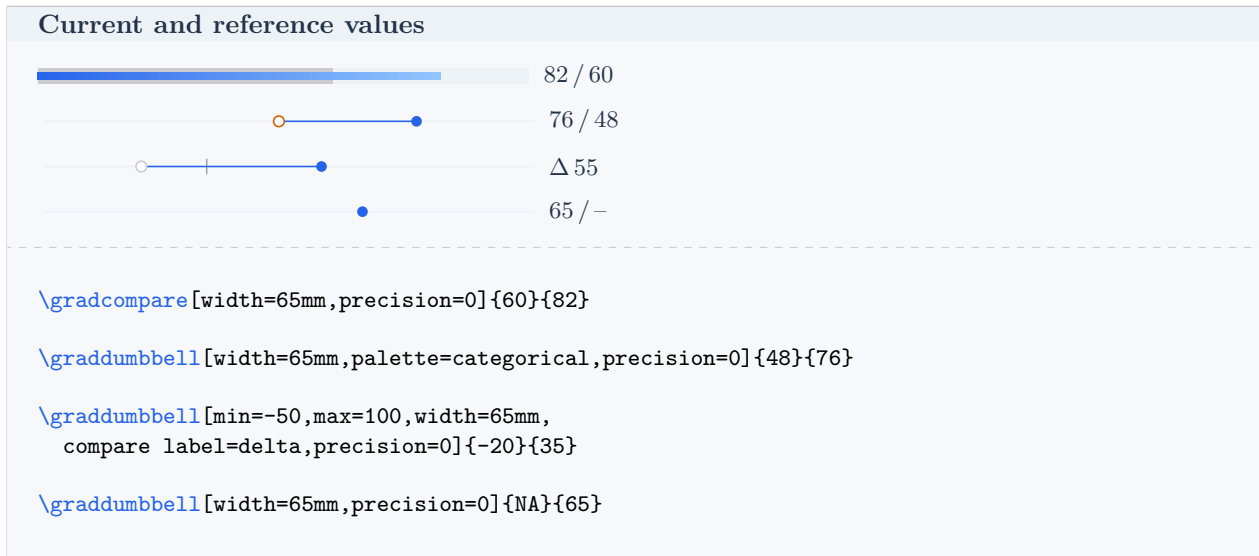
\gradrange[min=-50,max=100,width=65mm,range point=10]{-20}{35}

```

Error magnitudes are nonnegative distances from the observation, not absolute endpoints. Explain whether they represent standard deviation, standard error or confidence intervals in the table caption. Targets and bands are references, not automatically estimated uncertainty. A band must have ordered endpoints within the scale; invalid reference bands and targets cause errors.

Floating intervals require lower $\leq$ upper; equal endpoints draw a cap, not an inflated bar. Both endpoints must be present or both missing. A range point must lie inside the original endpoints. Intervals reject error whiskers and quality thresholds. Default labels show the supplied endpoints. Lollipops show a point for zero, but no point for missing input.

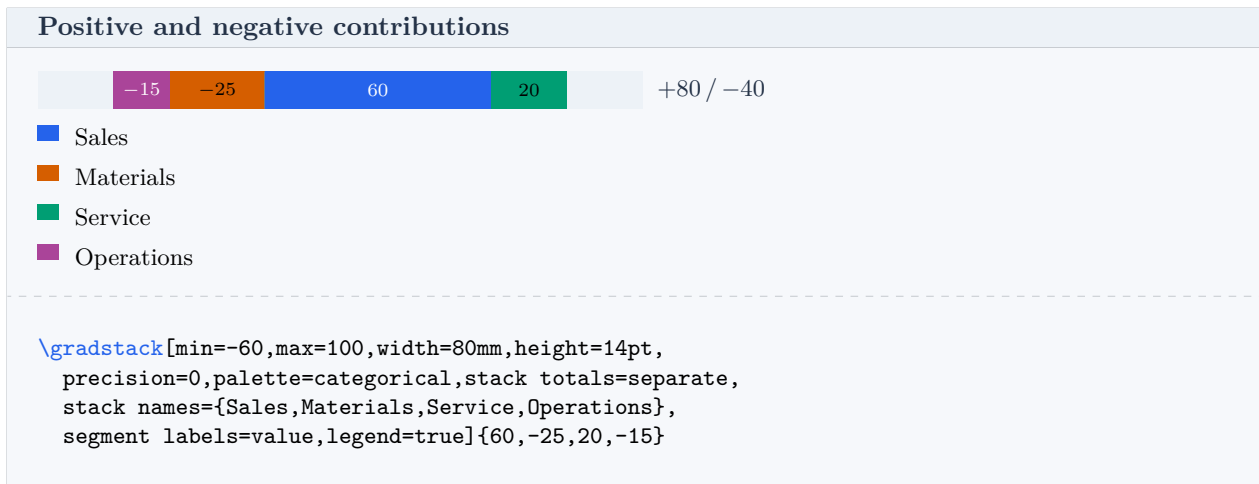
6 Layered and dumbbell comparisons



Layered bars use a wide reference and a centered current layer. The `compare ratio` defaults to .45 and must be strictly between zero and one. Both lengths start from the common zero; they are not added or divided.

Dumbbells use a hollow reference point and a current point joined by a line. The default label is current / reference; delta means current minus reference. A missing endpoint suppresses the connector; delta is missing if either endpoint is missing. Equal endpoints overlap. Error whiskers refer to the current value. `compare color`, `marker size` and `stem width` control appearance. Dedicated comparison and spark commands retain their shape when applying presets.

7 Stacked contributions and segment labels



Positive and negative contributions accumulate independently from zero. Within each side, segment order follows input order. Negative segments require a negative min. Positive and negative subtotals are checked separately against the scale. The default total is the algebraic sum; `stack totals=separate` shows positive and negative subtotals. A zero net does not erase nonzero contributions.

Segment labels can be none, value, percent, name, name value or name percent. The percentage of a segment is

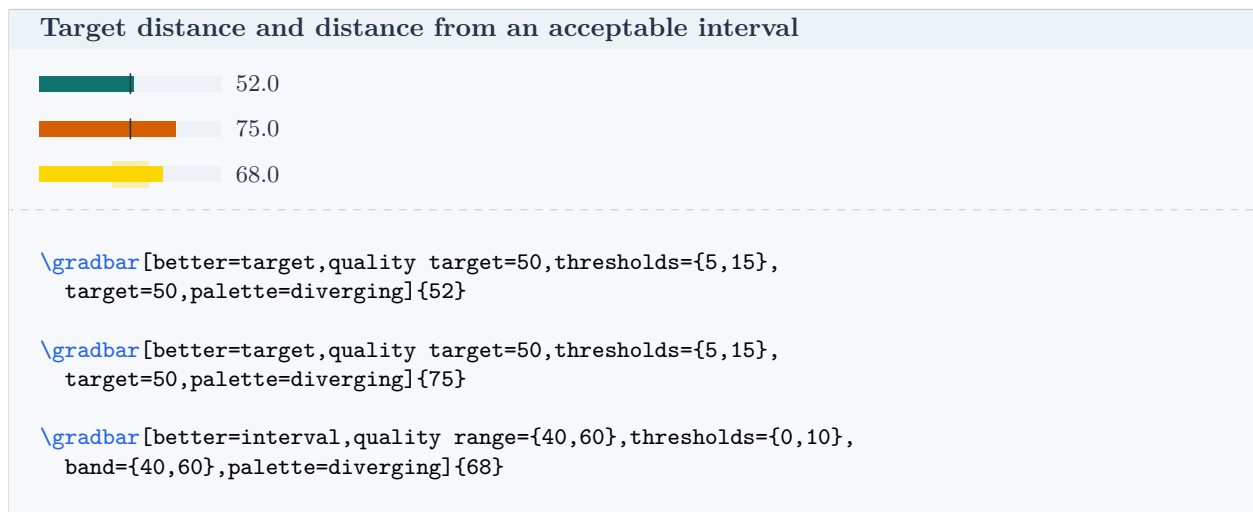
$$100|v_i|/\sum_j|v_j|,$$

including clipped contributions. These percentages represent shares of absolute activity, not shares of a potentially zero net. All-zero stacks do not calculate percentages. Stacks are not normalized to full width. Missing segments, empty lists, quality thresholds and error whiskers are rejected.

Provide one stack name per segment for named labels or automatic legends. Small labels move outside with leader lines; automatic black/white text follows the fill. Zero segments retain their palette and legend positions. Use a shared `\gradbarslegend` outside a table to avoid repeating the legend for every row.

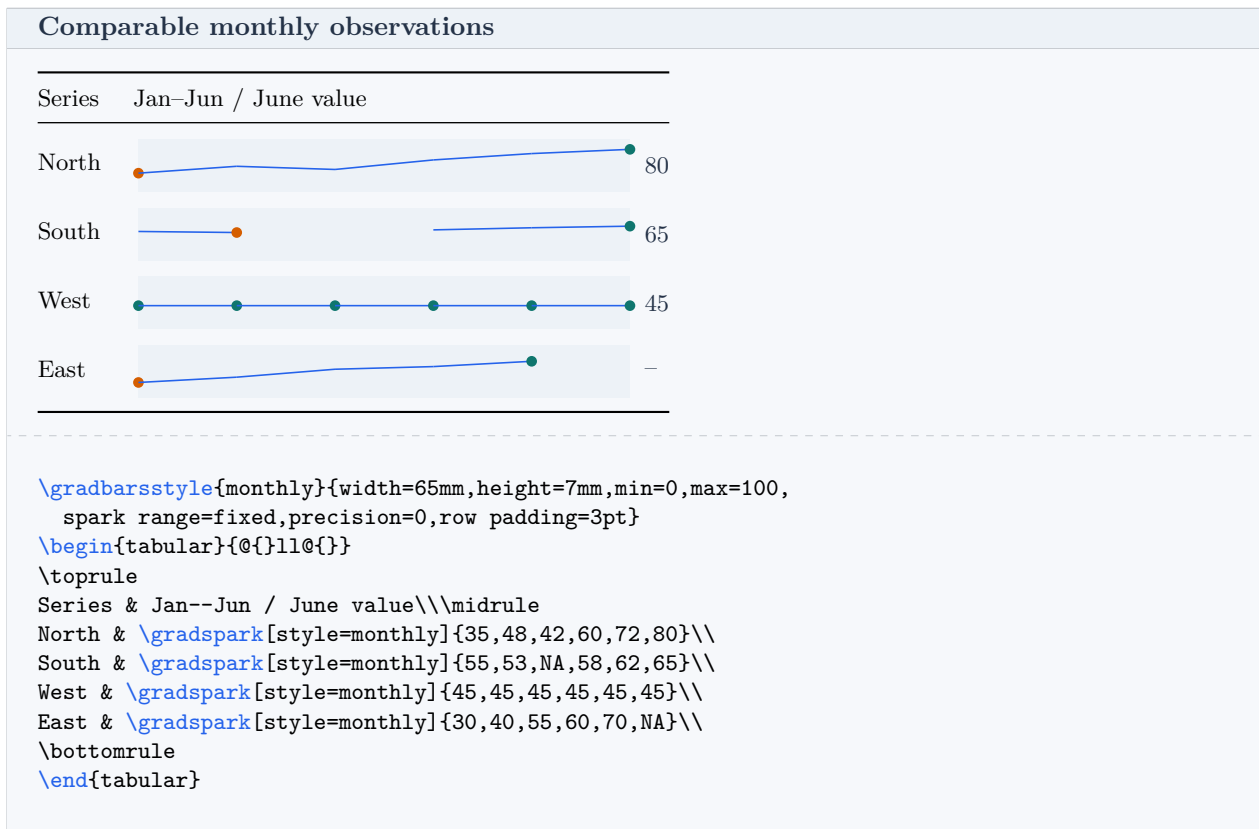
8 Quality: higher, lower, target or interval

Higher and lower modes classify the raw value using two increasing thresholds. Lower reverses the quality colors without reversing geometry. Threshold colors are always ordered bad, intermediate, good.



Target mode evaluates $d = |v - t|$. Interval mode evaluates $d = \max(0, a - v, v - b)$ for an ordered closed interval $[a, b]$. These modes require two nonnegative increasing distance thresholds: $d \leq t_1$ is good, $t_1 < d \leq t_2$ intermediate, and $d > t_2$ bad. Thus thresholds of 0 and 10 classify values inside the interval, including its endpoints, as good. Quality target/range evaluate data; target/band draw optional references. Set them separately. Quality changes color, never positions or labels. It is not applied to stacks, floating intervals or sparklines.

9 Sparklines and missing positions



Observations are equally spaced. Empty fields and NA keep their horizontal position and break connections. The last label is the last position, not the last nonmissing observation. Default height is 4ex.

spark range=auto uses each row's extrema; compare shapes only with this mode. Use fixed with shared min, max, width and height to compare magnitudes across rows. Constant auto series are centered vertically; a singleton is centered horizontally. All-missing series show only a track and missing label. Fixed ranges use normal overflow rules. Auto ranges do not support computed-percent labels.

spark points can be extrema (default), all, last or none. Extrema mode marks all tied extrema and the last position if present. Low and high colors default to gradbarsOrange and gradbarsTeal. A constant series uses the high color; a non-extreme last point uses the line color. Point colors indicate position, not quality. Use color, stem width and marker size to style the line and points. Sparklines do not parse dates and do not use bar targets, error whiskers, bands or thresholds. Labels stay outside. Use a common observation grid for irregular dates.

10 CSV input and numeric table columns

CSV files are read as characters, not executed as TeX. UTF-8 BOM, quoted commas, doubled quotes and custom missing tokens are supported. Fields must fit on a single line and the delimiter must be a comma. Headers must be unique and nonempty. The following source assumes a CSV file with Name and Score columns:

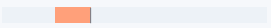
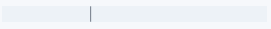
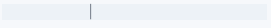
```

\gradbarsloadcsv{results}{results.csv}
\gradbarscsvtable{results}{Name}{Score}
\gradbarscsvstyle{shared}{results}{Score}
\gradbarcsv[style=shared]{results}{1}{Score}
\gradbarscsvcell{results}{1}{Name}

```

Rows are one-based. CSV range styles include zero and use a positive upper limit even for all-zero, all-negative or all-missing columns. Table generation does not automatically paginate. Keep your own longtable structure and use individual CSV cells when pagination matters. A dataset definition follows grouping.

Only numbers in a common-style column

Item	Value
A	 72
B	 -20
C	 0
D	 -

```

\gradbarscolumn{G}{width=35mm,min=-50,max=100,precision=0}
\begin{tabular}{@{}lG@{}}
\toprule
Item & \multicolumn{1}{l}{Value}\\\midrule
A & 72\\
B & -20\\
C & 0\\
D & NA\\
\bottomrule
\end{tabular}

```

Numeric columns use `colcell`. Use `multicolumn` for text headings. They do not calculate a range and are not siunitx S columns. Avoid placing complete drawing commands in a numeric collection cell. `tabular` and `longtable` workflows are documented; a `tabularray`-specific collection interface is not implemented.

11 Label placement, printing and practical limits

Labels support `outside`, `inside`, `end`, `auto` and `none`. `Auto` moves a label outside when it does not fit. `End` places it above the endpoint. Comparisons, lollipops, outlined bars and intervals with a range point move internal labels outside. Sparklines always use outside labels. `text color=auto` chooses black or white. `label width` reserves an outside slot; use the same sufficiently wide slot throughout a column. Long labels may require a wider column or a custom short text.

`row padding` reserves extra height above and below a graphic; it does not change the document's global row spacing. Use `arraystretch` or table row spacing when appropriate. Marker shapes reserve space consistently even when the first or last observation does not have a marker.

`print` applies gray fills, textures and white label backgrounds. `Mono` is a group palette using those settings. Provide textual category names and real values as well as visual distinctions. Outlined bars and legends are available through `outline=true` or the `outline` preset; `print` clears `outline` mode.

All collision handling is local to a graphic, not to the whole table. Complex axes, date parsing and large general-purpose plots are outside this package's scope. No specialized tagged-PDF chart descriptions are generated.

12 Compact option reference

Option	Default / meaning
width, height	24mm, 1.5ex; sparklines default to 4ex high.
min, max	0, 100; common scale.
precision, number format	1; fixed, grouped or scientific.
unit, text	Empty; suffix or complete custom label.
label, label width	outside, 4.5em.
rounded, row padding	0pt, 0pt; nonnegative lengths.
marker size, stem width	2pt, .6pt; positive lengths.
outline, outline width	false, .4pt; positive width.
compare color, compare ratio	black!20, .45; ratio strictly between 0 and 1.
compare label	values or delta.
stack totals	net or separate.
stack colors	gradbarsBlue, gradbarsTeal, gold; cyclic.
stack names, legend	Empty list; false.
segment labels	none, value, percent, name, name value, name percent.
better	higher, lower, target, interval.
quality target, quality range	Required for their respective modes; no default.
spark range, spark points	auto; extrema.
overflow	clip or error.

13 Contact and maintenance

The author and maintainer is SuFan, whose GitHub account is Tape5try. Contact 3546236610@qq.com or report an issue at <https://github.com/Tape5try/gradbars/issues>. Include a minimal source, engine name, TeX distribution and log. The MIT license is included in the distribution. Third-party dependencies retain their own authorship and licenses; they are not part of this package's copyright claim.